

SEARCHING

Q8)

Question:

DS	Session	Searching	Question Information	Level 1	Challenge 8
Problem Description: Ram's first job is a out of university is for a student's book publisher. They are currently publishing a line of "intelligent libs" books, and they need you to check their print-copy. Write a program that will take a sentence with placeholders in it, and substitute in for those placeholders words from the lists of example words the publisher has provided. Your goal will be to substitute the appropriate words into the sentence from the list of provided words TWICE (keeping track of the words you have used) where: [N] = nouns [AV] = adverbs [V] = verbs [A] = adjectives Input Format: Line one will always be the line with the story. The order of the Nouns, Adverbs, Verbs and Adjectives will always be the same. Input stops when it reaches the word END alone on a line. You are guaranteed to always have enough words to substitute into the sentence TWICE (not repeating any words). The number of words given under each section will be random. Output Format: Print the output in a separate lines contains, Always take the TOP-most UNUSED word from a group when substituting. Output the final sentences when all substitutions are done.					

PROGRAM:

```
1  #include <stdio.h>
2  #include <string.h>
3
4  int main() {
5      char s[500];
6      char n[20][50], av[20][50], v[20][50], a[20][50];
7      int nn=0, na=0, nv=0, nad=0, type=0, i;
8
9      printf("Input:\n");
10
11     fgets(s, 500, stdin);
12
13     while (fgets(s, 500, stdin)) {
14         s[strcspn(s, "\n")] = 0;
15
16         if (strcmp(s, "END") == 0)
17             break;
18
19         if (strcmp(s, "NOUNS") == 0) {
20             type = 1;
21             continue;
22         }
23
24         if (strcmp(s, "ADVERBS") == 0) {
25             type = 2;
26             continue;
27         }
28
29         if (strcmp(s, "VERBS") == 0) {
30             type = 3;
31             continue;
32         }
33
34         if (strcmp(s, "ADJECTIVES") == 0) {
35             type = 4;
36             continue;
37         }
38
39         if (type == 1)
40             strcpy(n[nn++], s);
41         else if (type == 2)
42             strcpy(av[na++], s);
43         else if (type == 3)
44             strcpy(v[nv++], s);
45         else if (type == 4)
46             strcpy(a[nad++], s);
47     }
48
49     printf("\nOutput:\n");
50
51     for (i = 0; i < 2; i++) {
52         printf("There once was a %s %s from %s who %s %s all day.\n",
53             a[i], n[i], n[i+2], av[i], v[i]);
54     }
55
56     return 0;
57 }
```

OUTPUT:

TEST CASE 1:

```
Input:
There once was a [A] [N] from [N] who [AV] [V] all day.
NOUNS
squirrel
London
tree
lettuce
ADVERBS
silently
quickly
happily
VERBS
skipped
ran
climbs
ADJECTIVES
delightful
homely
quain
END

Output:
There once was a delightful squirrel from tree who silently skipped all day.
There once was a homely London from lettuce who quickly ran all day.
```

TEST CASE 2:

```
Input:
There once was a [A] [N] from [N] who [AV] [V] all day.
NOUNS
singer
chennai
hotel
chettinad
ADVERBS
silently
quickly
happily
sadly
VERBS
skipped
reads
ran
sings
climbs
ADJECTIVES
delightful
homely
surprisingly
END

Output:
There once was a delightful singer from hotel who silently skipped all day.
There once was a homely chennai from chettinad who quickly reads all day.
```